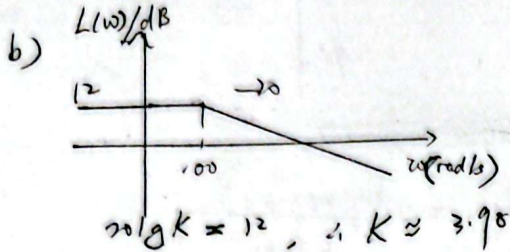


$\therefore 20 \lg K = 60, \therefore K = 10^3$

$$G(s) = \frac{10^3 (\frac{1}{400}s + 1)}{(\frac{1}{2}s + 1)(\frac{1}{200}s + 1)(\frac{1}{4000}s + 1)}$$

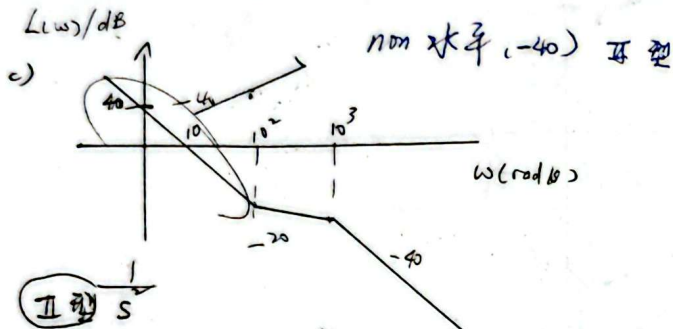
(rad/s)

$$\begin{aligned} \omega_1 &= 2 & T_1 &= \frac{1}{2} \\ \omega_2 &= 200 & T_2 &= \frac{1}{200} \\ \omega_3 &= 400 & T_3 &= \frac{1}{400} \\ \omega_4 &= 4000 & T_4 &= \frac{1}{4000} \end{aligned}$$



$$G(s) = \frac{K}{T_1 s + 1} = \frac{3.98}{\frac{1}{100}s + 1}$$

$T_1 = \frac{1}{\omega} = \frac{1}{100}$



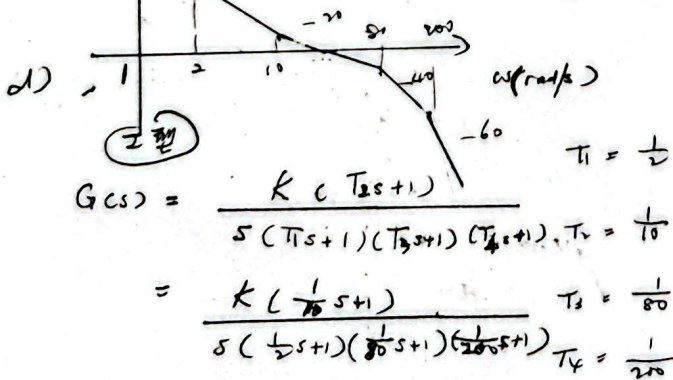
$$G(s) = \frac{K (T_1 s + 1)}{s^2 (\frac{1}{5}s + 1)}$$

$T_1 = \frac{1}{\omega_1} = \frac{1}{100}$
 $T_2 = \frac{1}{\omega_2} = \frac{1}{1000}$

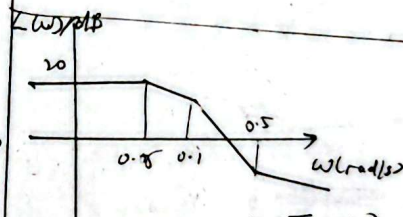
$$= \frac{100 (\frac{1}{100}s + 1)}{s^2 (\frac{1}{1000}s + 1)} \quad K = \angle(\omega=1 \text{ rad})$$

$$= \frac{-(\angle(\omega=1) - 0)}{\frac{10}{10}(\text{dec})} = -40$$

$\therefore \angle(\omega=1) = 40$
 $\therefore K = 10^2$



$\angle(\omega=1) = 40 = 20 \lg K$
 $\therefore K = 100$



e)

$$G(s) = \frac{K (T_2 s + 1)}{(T_1 s + 1) (T_3 s + 1)}$$

$T_1 = \frac{1}{0.05}$
 $T_2 = \frac{1}{0.1}$
 $T_3 = \frac{1}{0.5}$

$$= \frac{10 (2.5s + 1)}{(20s + 1) (10s + 1)}$$

$20 \lg K = 20$
 $K = 10$

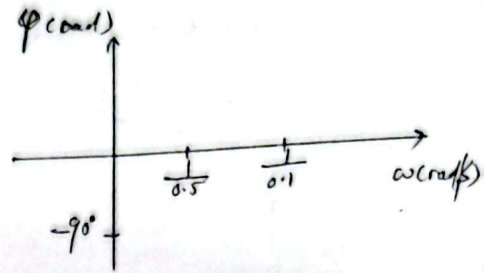
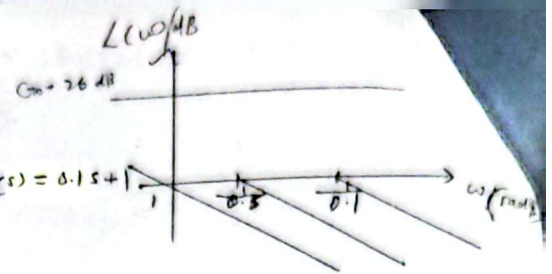
$$Q2 \quad (1) \quad G(s) = \frac{20}{s(0.5s+1)(0.1s+1)}$$

Ex. Bode

$$K = G_0(s) = 20, \quad G_1(s) = \frac{1}{s}, \quad G_2(s) = \frac{1}{0.5s+1}, \quad G_3(s) = \frac{1}{0.1s+1}$$

$$L(\omega) = 20 \lg 20 \approx 26 \text{ dB}$$

$$\phi(\omega) = -20^\circ = 0^\circ$$



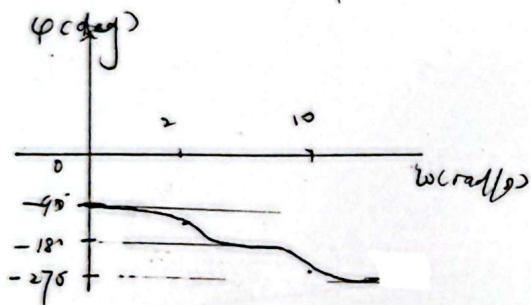
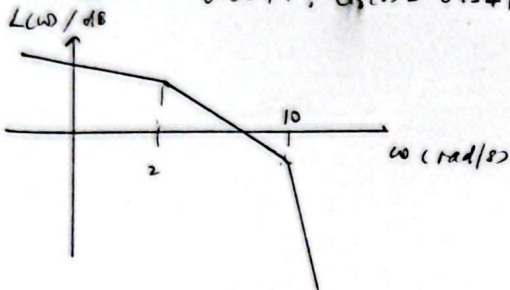
$$G(s) = T_1 s + 1$$

$$T = \frac{1}{\omega}, \quad \omega = \frac{1}{T}$$

$$Q2 \quad (1) \quad G(s) = \frac{20}{s(0.5s+1)(0.1s+1)}$$

$$G_0(s) = 20 \lg k = 20, \quad G_1(s) = \frac{1}{s}, \quad \therefore K = 26 \text{ dB}$$

$$G_2(s) = 0.5s + 1, \quad G_3(s) = 0.1s + 1$$

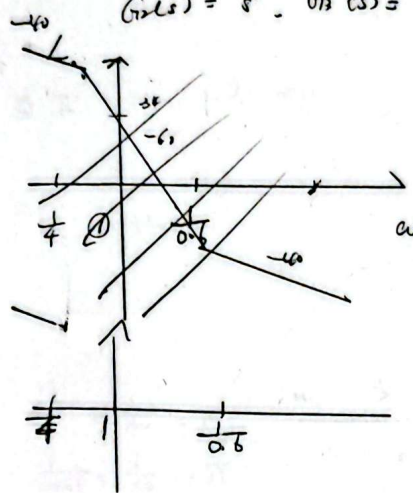


$$G(s) = \frac{50}{s^2(0.6s+1)}$$

$$(3) \quad G(s) = \frac{50(0.6s+1)}{s^2(0.6s+1)}$$

$$G_0(s) = 50, \quad K = 20 \lg 50 \approx 34, \quad G_1(s) = 0.6s + 1$$

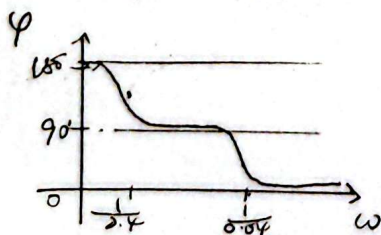
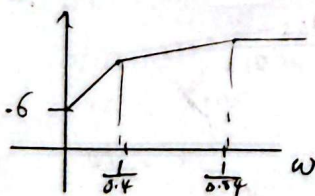
$$G_2(s) = s^2, \quad G_3(s) = 48 + 1$$



$$(2) \quad G(s) = \frac{28}{(0.04s+1)(0.04s+1)}$$

$$G_0(s) = 28, \quad G_1(s) = 0.04s + 1, \quad \therefore K = 20 \lg 28 = 6 \text{ dB}$$

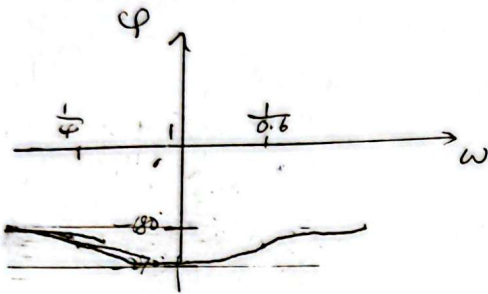
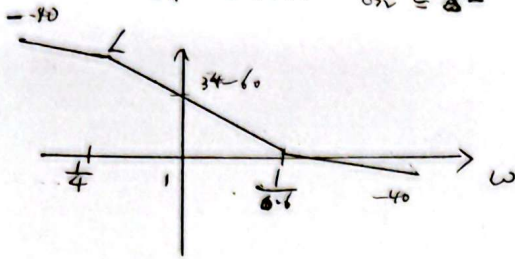
$$G_2(s) = 0.4s + 1, \quad G_3(s) = 0.04s + 1$$



$$Q2(3) \quad G(s) = \frac{30(0.6s+1)}{s^2(4s+1)}$$

$$G_0(s) = 30, \quad K = 20 \lg 30 \approx 34 \text{ dB}$$

$$G_1 = 0.6s+1 \quad G_2 = \frac{1}{s^2} \quad G_3 = 4s+1$$



$$Q2(4) \quad G(s) = \frac{7.5(0.2s+1)(s+1)}{s(s^2+16s+100)}$$

$$= \frac{\frac{3}{40}(0.2s+1)(s+1)}{s(\frac{s^2}{100} + \frac{16s}{100} + 1)}$$

$$G_0(s) = \frac{3}{40}, \quad K = -22.5 \text{ dB}$$

$$G_1 = 0.2s+1, \quad G_2 = s+1$$

$$G_3 = \frac{1}{s}, \quad G_4 = \frac{s}{100} + \frac{16s}{100} + 1$$

